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Research paper

Scaling-based design approach for tailbeat fish-inspired robots swimming at high speed and propulsive efficiency

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A B S T R A C T

The synchronous integration of multiple parameters in developing high-speed robotic fish remains a challenge. Consequently, most existing fish-like robots have been created through trial-and-error methods. This paper presents a comprehensive scaling-based design approach for developing a 0.32-m-long robot, KUFish-II, capable of achieving a target speed of 1.75 m/s or 5.5 body lengths per second (BL/s). This speed was predicted based on three physical parameters: Froude number, Strouhal number, and reduced frequency. To design a robot capable of achieving the speed, the proposed scaling law specified the stroke amplitude, fin area, tailbeat frequency, and torque needed for required thrust generation and high swimming efficiency. Subsequent experiments showed good agreement between the measured and predicted thrusts, frequencies, and swimming speeds. Swimming at a maximum speed of 1.79 m/s (5.6 BL/s), with a low Strouhal number and a narrow peak-to-peak amplitude of the tail, KUFish-II can be regarded as one of the few fast-swimming robots capable of achieving efficient locomotion with a low cost of transport. These results confirm the reliability of the proposed approach and highlight its potential use for designing efficient, large-scale, fast-swimming robotic fish.

1. Introduction

Fish locomotion has been extensively explored over the past several decades (Salazar et al., 2019; Sfakiotakis et al., 1999; Webb, 1988). Despite significant technological advancements, fully understanding the swimming mechanics of fast-swimming oceanic species (Ren and Yu, 2021; Walters and Fierstine, 1964) remains a challenge. Thus, fast-swimming fish have long inspired researchers seeking to replicate their high-speed aquatic movements in robotic fish designs (Anderson and Chhabra, 2002; Clapham and Hu, 2014, 2015; Hu et al., 2015; Iguchi et al., 2024; Pham et al., 2023; Tsimbo Fokou et al., 2023; Wang et al., 2010; Wen et al., 2013; White et al., 2021; Yu et al., 2016; Zhong et al., 2017, 2021; Zhu et al., 2019). Investigating fast-swimming behaviors can provide valuable insights that support the development of robotic fish for practical applications (Raj and Thakur, 2016). To achieve high speed and fast acceleration, fish typically undulate their bodies and beat the caudal fin (CF), which is known as the BCF mode, rather than using the median and/or paired fins (MPF mode), which generate relatively smaller thrust, and thus result in slower swimming (Breder, 1926; Sfakiotakis et al., 1999). Some fish can reach swimming speed exceeding 20.0 body lengths per second (BL/s) while exhibiting a wide range of tailbeat frequencies from 0.1 to 20.0 Hz (Bainbridge, 1958). For example, the yellowfin tuna, a thunniform fish (Shadwick and Syme, 2008), can reach a maximum speed of 20.8 m/s (21.1 BL/s) (Walters and Fierstine, 1964).

For real ocean applications, a robotic fish must be able to swim forward to overcome a stream under the sea whose speed can reach 2.5 m/s near the sea surface (Driscoll et al., 2008). As a result, the maximum swimming speed of a robotic fish should be higher than the stream speed under the sea. However, current robotic fish prototypes have yet to match the swimming speeds of real fish (see Table 1). Despite efforts to reproduce thunniform swimming, a form of efficient and low-drag undulatory motion with increasing flexibility toward the posterior, Tunabot can achieve a maximum tailbeat frequency of 15.0 Hz but only reaches a swimming speed of about 1.0 m/s (Zhu et al., 2019). To the best of our knowledge, the carangiform-inspired robot iSplash-II is recognized as the fastest tailbeat robot, achieving a maximum speed of 3.7 m/s (11.6 BL/s) at a tailbeat frequency of 20.0 Hz (Clapham and Hu, 2015). It is true that achieving high-speed swimming in a robotic system is actually challenging and demanding a comprehensive understanding of aquatic locomotion, body shape design, and tailbeat patterns (Dean and Bhushan, 2010; Rossi et al., 2011; Zhong et al., 2021). These analyses require examining key factors, including tailbeat frequency, tail stroke angle, tail flexibility, tail stiffness, and tail fin area.

While making efforts to achieve high-speed swimming, improving efficiency is another common concern (Bartol et al., 2003; Dean and Bhushan, 2010; Jaya and Kartidjo, 2022; Lindsey, 1978; Matta et al., 2020; Yu et al., 2020). Extensive studies have been conducted on practical applications aimed at reducing body drag, including the replication of the highly efficient thunniform mode characterized by

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small undulatory angles of the anterior body (Hu et al., 2015; Matta et al., 2020; Shadwick and Syme, 2008; Zhu et al., 2019); the development of an energy-efficient ostraciiform-inspired robotic swimmer (Bartol et al., 2003); the implementation of riblet structures that mimic shark skin to reduce frictional drag in turbulent flow regimes (Dean and Bhushan, 2010); the equipment of median fins achieving up to 20% drag reduction (Liu et al., 2017), and the adoption of a tuna-inspired fusiform body shape to minimize flow separation along the body (Lindsey, 1978). In addition, a study by (Saadat et al., 2017) showed that fast swimmers could attain efficient locomotion when two conditions are satisfied: a peak-to-peak amplitude of the caudal fin (A_m) remains within $(10-30)\%$ of the body length (L), and a Strouhal number, representing how fast the tail beats relative to forward swimming speed (Lauder and Tytell, 2005; Triantafyllou et al., 1991), stays within the range of 0.25-0.35. The authors also concluded that power consumption was minimized when swimming with a narrow A_m within the same range of $(10-30)\%L$. Among these above methods, configurations that combine a fusiform body shape, median fins, and small stroke angles offer a strong potential for achieving fast and efficient swimming. However, replicating fast swimming speed is challenging mainly due to motor torque and speed limitations. The motor mass is another design parameter because a robotic fish should achieve neutral buoyancy for agile swimming. A robot with a heavier motor requires a larger body for neutral buoyancy, producing a higher drag. Thus, selecting a lightweight motor with adequate speed and torque is key for achieving the required frequency and thrust, and for further reducing body drag (Youssef et al., 2022).

Unfortunately, we could not find articles describing a complete systematic design procedure for robotic fish that incorporates the key considerations and predictions of the target swimming speed, required tailbeat frequency and torque, motor selection, neutral buoyancy, and other related factors. Therefore, it appears that the existing fast-swimming robotic fish (Clapham and Hu, 2015; Iguchi et al., 2024; Pham et al., 2023; Tsimbo Fokou et al., 2023; Wang et al., 2010; Wen et al., 2010, 2013; White et al., 2021; Zhong et al., 2017; Zhu et al., 2019) have been developed through a trial-and-error approach. For a systematic design, scaling laws are able to provide valuable physical insights into thrust and efficiency (Triantafyllou et al., 2005), in comparison with the analytical methods (e.g., elongated-body theory by (Lighthill, 1970)) and the aerodynamic reduced-order model by (Brunton et al., 2013). For example, the scaling relationships for thrust, power, and efficiency of a rigid flapping foil undergoing heaving and pitching motions were proposed by (Floryan et al., 2017, 2018) based on lift-based and added-mass forces. The scaling laws, fully established through water-tunnel tests, revealed that thrust scales quadratically with the foil-flapping frequency and the foil peak-to-peak amplitude, while scaling linearly with the propulsive area. A numerical study by (Yu and Huang, 2021) proposed a thrust-scaling relationship for NACA-type foils simulating the undulatory propulsion of anguilliform swimmers. The thrust scaling proposed by (Yu and Huang, 2021) depended on swimming speed, propulsive area, undulating wavelength, wave amplitude, undulatory frequency, as well as undulatory phase speed. The applicability of this thrust-scaling model was further demonstrated for carangiform swimmers. In general, for two models, the tail stroke angle was implicitly included in the peak-to-peak amplitude for (Floryan et al., 2017, 2018) and in the wave amplitude for (Yu and Huang, 2021). However, the thrust produced across different propulsive areas was not tested, leading to thrust scaling linearly with the propulsive area when two different propulsive areas are considered. More importantly, torque estimation, which is crucial for motor selection, was not addressed, highlighting a key limitation of the existing scaling laws in establishing a complete and systematic design methodology. To address these gaps, we proposed a scaling-based relationship capable of predicting the required thrust and torque for a target speed across different tailbeat frequencies, stroke angles, and tail fin area through a series of parametric studies to provide required frequency and torque for proper motor selection. Using physical parameters of Froude number (Fr), Strouhal number (St), and reduced frequency (σ), we first estimated the maximum swimming speed as the average of the speed calculated with a Fr of one and the two additional speeds estimated from the St and σ. To achieve this speed, scaling relationships were then applied to predict the key design parameters, such as the required thrust, torque, tailbeat stroke angle and CF area, enabling the robotic fish to swim at the target maximum speed and tailbeat frequency.

For validation of our proposed scaling-based approach, we developed a 0.32-m-long robotic fish, namely, KUFish-II, capable of achieving a target swimming speed of 1.75 m/s. A complete six-step design process, from literature survey to swimming verification, is illustrated in Fig. 1 (see Fig. A1 for the full development flowchart). The development of KUFish-II first followed the proposed design approach based on the reference parameters from our previously developed KUFish (Pham et al., 2023) and the target swimming speed, which was computed using literature-informed values of the Fr, St, and σ (Anderson et al., 1998; Bandyopadhyay, 2002; Belmonte et al., 1998; Eloy, 2012; Taylor et al., 2003). Additional parameters, including mass, volume, body wetted area, tail fin area, tailbeat amplitude, and required torque and thrust production, were predicted using this scaling-based relationship. In addition, the proposed design was validated through both tethered and free-swimming tests, including the measurements of frequency, thrust, power consumption, undulatory swimming kinematics, and free-swimming speed. The body drag coefficient was further computed by a computational fluid dynamics (CFD) simulation, and the swimming speed and distance were examined by a one-dimensional (1D) equation of motion (EoM). Thus, in this work, we have proposed

Table 1
Fish-inspired robots.

Group	Model	U (mL/s)	U (BL/s)	f (Hz)	Fr	St	σ	A_m	References
A	SoRoFAAM-1	0.05	0.2	1.00	0.03	0.63	35.3	0.11L	Tsimbo Fokou et al. (2023)
	Wen's carangiform	0.22	0.4	0.80	0.09	0.43	13.6	0.20L	Wen et al. (2013)
	VCUUV	1.25	0.5	1.00	0.26	-	12.1	-	Anderson and Chhabra (2002)
	ACPM fish robot	0.67	2.2	1.50	0.38	0.36	4.4	0.52L	Zhong et al. (2017)
	iSplash-I	0.70	2.8	6.60	0.45	0.41	14.9	0.17L	Clapham and Hu (2014)
	SPC-3 UUV	1.87	1.2	2.50	0.47	0.45	13.4	0.21L	Wang et al. (2010)
	Tunabot	0.82	3.2	10.00	0.52	0.40	19.6	0.13L	Zhu et al. (2019)
	Thunniform robot	2.00	1.7	2.80	0.59	-	10.4	-	Hu et al. (2015)
	Tunabot flex	1.17	4.6	8.00	0.74	0.46	10.9	0.26L	White et al. (2021)
	KUFish	1.40	6.4	11.35	0.95	0.45	11.2	0.25L	Pham et al. (2023)
B	KUFish-II	1.79	5.6	12.20	1.01	0.28	13.7	0.13L	Current robot
	Agile robotic fish	2.60	6.3	10.00	1.29	0.30	10.0	0.19L	Iguchi et al. (2024)
	iSplash-II	3.70	11.6	20.00	2.09	0.34	10.8	0.20L	Clapham and Hu (2015)

Group A: regular robotic fish.
Group B: fast, efficient, and low-input-power robotic fish.

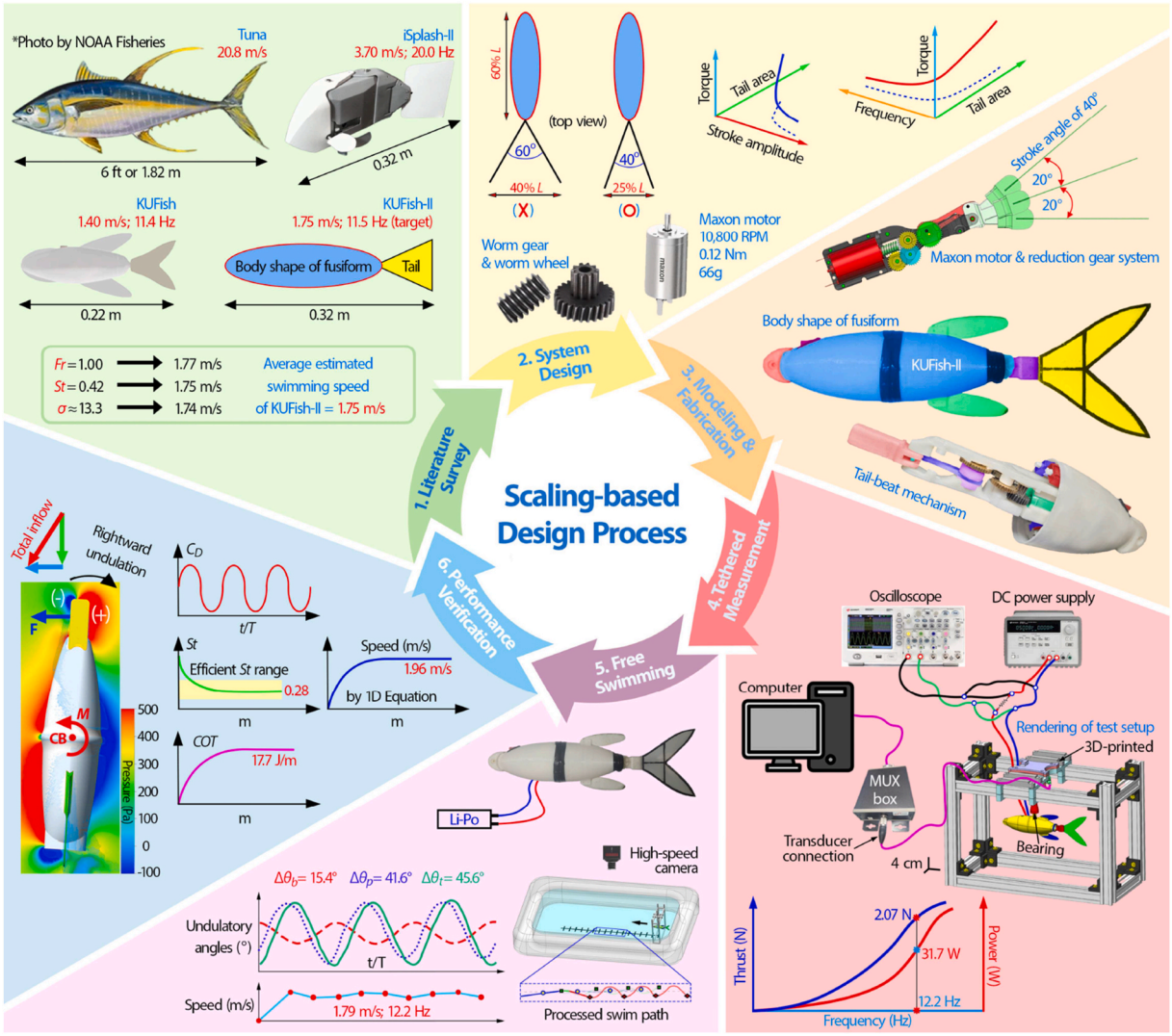


Fig. 1. Diagram of the six-step scaling-based design process for tailbeat robotic fish. The process begins with a literature survey to determine the target swimming speed. The system design phase searches for the suitable CF area and computes the required thrust and torque, tailbeat frequency, and tailbeat stroke amplitude. The fabrication step illustrates the CAD modeling and final KUFish-II model. Tethered and free-swimming experiments are conducted to validate the design, followed by the performance verification and analysis.

an experimentally proven systematic approach for designing fast and efficient fish-inspired swimming robots, which can potentially be used for large-scale, fast-swimming robotic fish.

2. Materials and methods

2.1. Scaling-based design approach for tailbeat mechanism

As a starting point, we consider the following question: How fast can the 0.32-m-long KUFish-II swim while maintaining high efficiency? To address this, we examine literature-informed values for three metrics commonly used to assess aquatic swimming performance: the Froude number, Strouhal number, and reduced frequency. The first consideration is the Froude number, defined as $Fr = U/(gL)^{0.5}$, where U is the maximum swimming speed (m/s) and g is the Earth's gravity (m/s^2). This metric is used to evaluate the characteristics of fluttering motions, describe the relationship between inertial and gravitational forces, and characterize the maneuverability of swimming. Previous studies (Bandyopadhyay, 2002; Belmonte et al., 1998) have shown that objects featuring long lightweight strips with oscillatory motion can achieve a Fr of up to 1.0. However, most robotic fish swim at $Fr < 1.0$, as shown in Table 1. Among them, the agile robotic fish and iSplash-II swam with Fr higher than those of the others ($Fr = 1.29$ and 2.09, respectively), indicating that their motion was consistently within the high-speed and burst-swimming regime (Bandyopadhyay, 2005).

Another prominent factor in analyzing swimming modes is the Strouhal number (St) (Triantafyllou et al., 1991), which is defined as $St = fA_m/U$, where f is the tailbeat frequency (Hz) and A_m is the peak-to-peak amplitude of the caudal fin. The maximum A_m of $0.2L$ is commonly observed in real fish, as reported by (Bainbridge, 1958; Saadat et al., 2017). Later studies have shown that both swimming and flying animals (Eloy, 2012; Taylor et al., 2003) generally achieve energy-efficient motion within the same range of $0.25 < St < 0.35$. For robotic fish, except for the agile robotic fish ($St \approx 0.3$), iSplash-II ($St \approx 0.34$), and ACPM fish robot ($St \approx 0.36$), most robotic fish swam at higher St than natural swimmers (see Table 1).

The third metric is the reduced frequency σ, which represents the acceleration reaction to hydrodynamic forces and relates the period of a water particle traversing along the body length to the period required for a complete tailbeat cycle (Webb, 1988). It is defined as $\sigma = 2\pi fL/U$. Typically, the reduced frequency for aquatic animals falls within the range of 0.1 to 0.4, which indicates both acceleration reaction and

hydrodynamic force components contributing to the force balance (Webb, 1988). However, for robotic swimmers, the reduced frequency far exceeds 0.4, typically ranging from 10 to 15 (see Table 1), which is significantly larger than that of natural species mainly due to their longer bodies and lower swimming speeds.

Our objective is to create a fast-swimming robot capable of reaching a *Fr* of 1.0. For a given body length (0.32 m) and literature-informed tail peak-to-peak amplitude ($A_m = 0.2L$), we selected the target tailbeat frequency ($f = 11.5$ Hz) and three dimensionless numbers (*Fr*, *St*, and σ), such that the corresponding three computed swimming speeds became comparable across cases (see the flowchart in Fig. A1; Appendix A). It was noticed that the selection of (*St* and σ), whose values should fall within the typical ranges, primarily relied on the reference maximum speed of 1.77 m/s, which was predicted by the *Fr* of 1.0. This speed stands for high-speed swimming or burst motion as $Fr = 1.0$ typically represents the onset of burst motion in aquatic species (Bandyopadhyay, 2005). The swimming speed computed using *Fr* depends solely on body length, while *St* and σ require two other design parameters: (f and A_m) for *St*, and (f and L) for σ. The resulting estimated speeds were 1.75 m/s (using $St = 0.42$) and 1.74 m/s (using $\sigma = 13.3$). These three estimated swimming speeds suggested an average target swimming speed of 1.75 m/s (5.5 BL/s). See the predicted values in Table B1 for details. Nevertheless, this raises a question about the effect of choosing a tailbeat frequency different from 11.5 Hz, such as 5.0 or 20.0 Hz. A detailed discussion on selecting the target frequency is presented in Appendix C.

Here, we describe a scaling-based design approach to determine design parameters, such as the body surface area, volume, and mass of the KUFish-II. All the reference parameters of the KUFish (Pham et al., 2023) and the predicted data of the KUFish-II are summarized in Table B2. The reference parameters of the KUFish were scaled to predict those of the current KUFish-II design to ensure that a target speed of 1.75 m/s is achieved. More specifically, the robots were assumed to reach their maximum swimming speed and maintain constant speeds after $m\dot{U} = 0$, indicating that the thrust (T) and body drag (D) were balanced, as expressed by the following 1D equation:

$$T - D = m\dot{U}. \tag{1}$$

In Eq. (1), the body drag is defined as $D = 0.5\rho U^2 S C_D$, where ρ is the water density, S is the body surface area, and C_D is the drag coefficient during undulatory swimming. We assumed that the C_D of the KUFish-II remains identical to that of the KUFish ($C_D = 0.032$) (Nguyen et al., 2022), whereas owing to the increase in motor mass, its assumed body mass (M_2) is 20% greater than that of the KUFish (M_1), corresponding to $M_2 = 1.2 \times M_1$. Consequently, for neutral buoyancy, the volume (V_2) of the KUFish-II should be increased by a factor of 1.2, and therefore, its body surface area (S_2) increases by $(1.2)^{2/3}$, compared with the corresponding V_1 and S_1 of the KUFish (Table B2). Using Eq. (1) when $\dot{U} = 0$ and the definition of drag, assuming the same C_D for the KUFish and KUFish-II, whose body and CF shapes are similar (Steps 1 and 3, Fig. 1) and aspect ratios are identical to 2.56 (computed by $AR = b^2/A$ (Sfakiotakis et al., 1999) in Table B2), the required thrust, T_2, of the KUFish-II is 1.91 N, which was calculated using Eq. (2) to achieve the predicted swimming speed, U_2.

$$T_2 = T_1 \times \frac{S_2}{S_1} \times \left(\frac{U_2}{U_1}\right)^2, \tag{2}$$

where the symbols with the subscripts "1" and "2" indicate the known parameters of the KUFish and the predicted parameters of the KUFish-II, respectively. Note that U_2 is the estimated target swimming speed (1.75 m/s) based on three physical parameters. To account for the relationship between the ratios of thrust (T_2/T_1) and the ratios of the CF area (A_2/A_1), tailbeat frequency (f_2/f_1), and tailbeat amplitude (θ_2/θ_1), Eq. (3) was derived from CFD results (Nguyen et al., 2023) as follows:

$$\frac{T_2}{T_1} = \left(\frac{A_2}{A_1}\right)^2 \times \left(\frac{f_2}{f_1}\right)^2 \times \left(\frac{\theta_2}{\theta_1}\right)^{2.15}, \tag{3}$$

where θ_2 is the desired tailbeat amplitude of the KUFish-II. While the same exponents for fin area and frequency in Eq. (3)–(5) were extracted based on the equations used to derive the added mass method (Pham, 2022), the exponents for tailbeat stroke amplitude in Eq. (4) and (5) were identified by the presented parametric studies based on CFD and added mass method in Appendix D. Note that the exponents for stroke amplitude in Eq. (4) and (5) represent average values from the parametric studies. Subsequently, for a selected A_2/A_1 and the computed ratios of thrust, frequency, and tailbeat amplitude, the predicted frequency f_2 was calculated using Eq. (4), as follows:

$$f_2 = f_1 \times \frac{A_1}{A_2} \times \sqrt{\frac{T_2}{T_1} \times \left(\frac{\theta_1}{\theta_2}\right)^{2.15}}. \tag{4}$$

Notably, A_2/A_1 should be selected such that the predicted f_2 from Eq. (4) closely matches the assumed or target frequency of 11.5 Hz, which is used to estimate the speeds based on the *St* and σ parameters (Step 1, Fig. 1). Then, this predicted f_2 is an input to estimate the required torque Q_2 in Eq. (5). For the calculation, we update the existing relationship of (Q_2/Q_1) as a function of (A_2/A_1) and (f_2/f_1), developed in (Nguyen et al., 2023), to account for the effect of change in stroke amplitude on torque using the added mass method in (Pham, 2022), as follows:

$$\frac{Q_2}{Q_1} = \left(\frac{A_2}{A_1}\right)^{2.5} \times \left(\frac{f_2}{f_1}\right)^2 \times \left(\frac{\theta_2}{\theta_1}\right). \tag{5}$$

Using Eq. (5), the required torque Q_2 can be estimated by the selected A_2/A_1, predicted f_2, and desired θ_2. The data used to extract Eq. (3) and (5) are summarized in Tables D1 and D2, respectively, with detailed explanations provided in Appendix D. For motor selection, the predicted f_2 in Eq. (4) and the required torque Q_2 in Eq. (5) are compared with the available frequency $\left(f_{2,spec}\right)$ and torque $\left(Q_{2,spec}\right)$, respectively, computed from the motor specifications: the nominal speed (revolutions per minute, *RPM*) in Eq. (6) and the stall torque (Q_s) in Eq. (7).

$$f_{2,spec} = \frac{RPM}{60 \times GR}, \tag{6}$$

$$Q_{2,spec} = Q_s \times GR, \tag{7}$$

where *GR* is the reduction gear ratio. The process of design parameter analysis is detailed in Section 3.1.

2.2. Design and fabrication

The body of the KUFish-II contains eight key components: anterior and posterior body parts, an additional nose fin (NF) section for neutral buoyancy, a tailbeat mechanism, a support structure for the mechanism, a motor, two rear fins (RFs), and a CF (see Fig. 2A–D). The external body shape was inspired by a fusiform design (Lindsey, 1978). Except for the motor, reduction gear system, and CF, most components, including the BF, supporting frames, NF, and RFs, were fabricated from polylactic acid (PLA) using a 3D printer (Cubicon, South Korea). Thus, its body is rigid and stiffer than the body of tuna, which has numerous spinal joints that allow for greater bending. The CF was fabricated by cutting a 0.6-mm-thick glass/epoxy plate.

A four-bar linkage was used to transform the motor rotational motion into the reciprocating motion of the peduncle, where the CF was attached through the rigid joint (Fig. 2D). In Fig. 2A, the lengths of the coupler O_1O_2 and crank O_2O_3 required to create a tailbeat angle θ were calculated using the predetermined lengths of the linkages OO_1 and OO_3 by the equations reported in (Pham et al., 2023). In Fig. 2A–C, the DCX motor was secured between the upper and lower 3D-printed frames, and

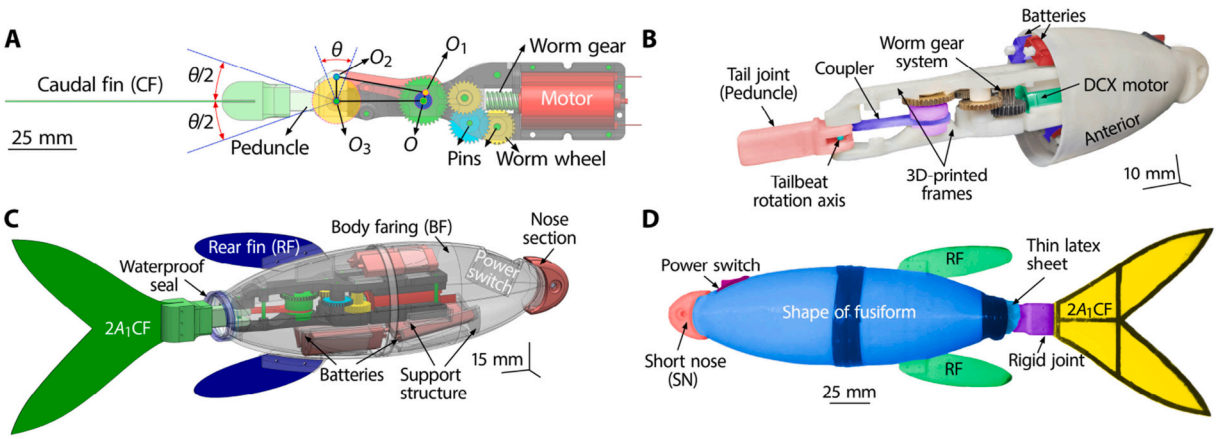


Fig. 2. Design and fabrication of KUFish-II. (**A-B**) Tailbeat mechanism: (A) rendering and (B) fabricated version with anterior fairing. (**C-D**) Fairing and CF: (C) rendering and (D) fabricated version. Note that the dimension annotations represent the unit scale bars for the 2D views (in A and D) and 3D views (in B and C), rather than the actual dimensions of the illustrated components.

seven Li-Po batteries were connected in series to form a 26-V battery package. A worm gear and multiple reduction gears were used to organize the transmission with a specific gear ratio. The actual operation of the tailbeat mechanism is shown in Movie S1. The center of mass of the robot was experimentally aligned with the CB and positioned below the body centerline for passive roll stability (Aureli et al., 2010; Pham et al., 2023). To prevent unwanted yaw and enhance the formation of LEV on the CF, the RFs were attached (Fig. 2C and D). With these arrangements, the KUFish-II can stably perform fast swimming, even without feedback control, and reach its maximum speed within a short time and distance. Waterproofing was achieved through the use of spray coating, waterproof tape, latex sheets, and epoxy.

2.3. Measurements of thrust and power consumption

The experimental setups for the frequency, thrust, power, kinematics, and swimming speed measurements are shown in Fig. 3A–C. In the tethered experiments (Step 4 in Figs. 1 and 3A), tailbeat motions were excited by an external DC power supply (Keysight E36103A, USA). The thrust generated by the CF was transmitted to a force transducer (ATI six-axis force/torque transducer, Gamma, SI 130-10, IP68, USA, force resolution of 0.025 N) via a 3D-printed coupling structure (Fig. 3A). The body could rotate symmetrically around the coupling structure using a stainless-steel ball bearing (C-SE608ZZ, Misumi, Japan). The force transducer was installed above the water surface and housed in a 3D-printed fixture mounted on a frame made of 40 cm × 40 cm aluminum profiles (Fig. 3A). The tailbeat frequency was measured by analyzing the recorded force signals.

The power consumption of the tailbeat mechanism was measured by calculating the product of the current passing through the motor and applied voltage. The current applied to the motor was measured using a resistor (R) and oscilloscope (TDS 2024; Tektronix Inc., USA; Step 4 in Fig. 1). Recorded within a 6-s display window, the voltage was sampled at 100 Hz for 1.5 to 2.5 s, corresponding to 15 to 25 tailbeat cycles. Then, the instantaneous power consumption at time t, $P(t)$, is computed using the instantaneous of the output voltage of the power supply, $Vol_1(t)$, and input voltage to the robot, $Vol_2(t)$, as follows:

$$P(t) = Vol_2(t) \times I(t) = Vol_2(t) \times \frac{Vol_1(t) - Vol_2(t)}{R}, \tag{8}$$

where $I(t)$ is the instantaneous current, and $Vol_1(t) - Vol_2(t)$ is the voltage across the resistor at time t.

Fig. 3B shows the setup for measuring the swimming speed and undulatory kinematics. The experiments were conducted in a pool measuring 2.5 m × 1.8 m × 0.4 m in length × width × height. A high-speed camera (HSC) (FASTCAM Mini UX50, Photron, Japan), which captured consecutive images with a pixel resolution of 1280×1024 at 250 Hz, was mounted 2 m above the floor. To minimize the optical issues of capturing images across air-water media, the swimming speeds were measured within the focal area ranging from 0.6 to 1.2 m after the robot departs (Fig. 3B). A release mechanism, which was mounted on a 20 cm × 20 cm aluminum frame positioned along the pool centerline, held the robot and ensured a consistent posture at the start of swimming (Fig. 3B). The frame was adjusted to submerge the robot at the mid-plane of the water, approximately 0.2 m below the water surface. A dummy mass was added to the nose section (Fig. 2C and D) to ensure neutral buoyancy and static balance. The recorded undulatory swimming was analyzed in MATLAB (MathWorks Inc., Natick, MA, USA) using the DLTdv-8 tool (Hedrick, 2008).

To validate the design of the tailbeat mechanism, the midline kinematics of the KUFish-II were analyzed. In Fig. 3C, the 2D pixel-based trajectories of six markers along the symmetric body axis were tracked: #1 at the tailbeat rotation axis, #2 at the nose, #3 at the body midpoint, #4 at the point connecting the peduncle and CF, #5 at the upper tip, and #6 at the lower tip. Distinguishing the upper and lower tip points of the CF is necessary because of the body pitching motion. The swim path of the KUFish-II is defined by the initial and final tracked positions of marker #3 because of its minimal lateral fluttering during swimming. Therefore, the angle between the swim path and the body axis is defined as the body's undulatory angle θ_b. Because of the CF deformation, the full-stroke angle of the peduncle, θ_p, which is defined as the angle between the body axis (connecting markers #1 and #2) and the peduncle axis (connecting markers #1 and #4), is regarded as the designed stroke amplitude of the tailbeat mechanism, θ (Fig. 2A). Meanwhile, the angle formed by the line connecting marker #4 and markers #5 or #6 at the tip of the CF and the peduncle axis (an extended line connecting markers #1 and #4, Fig. 3C) is defined as the tail bending angle, θ_t. In this case, θ_t may vary depending on which marker (#5 or #6) is used to measure the angle because the motion of the two markers is not identical. Thus, tracking markers #5 and #6 were necessary. The defined bending angle θ_t is the average of the two angles computed using these two markers.

3. Results

3.1. Design parameters

A design parameter study for the KUFish-II using Eq. (1)–(5) is illustrated in Fig. 4A. Specifically, to produce the same required thrust, the required torque (Q_2) was examined for different CF area ratios

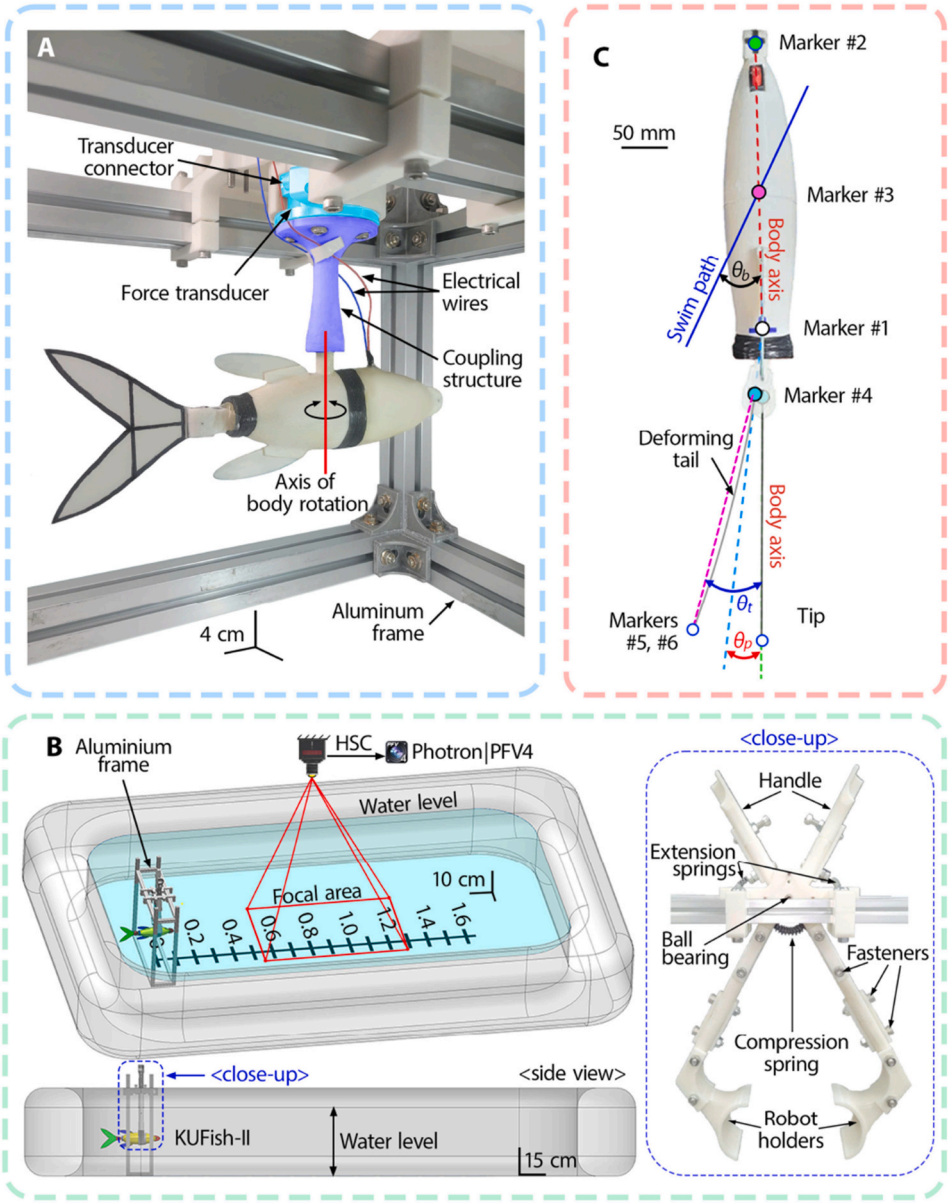


Fig. 3. Measurements of force, power, swimming speed, and undulatory kinematics. (**A**) Actual setup for the tethered experiments measuring force and power consumption (Step 4 in Fig. 1 for the CAD image). (**B**) Rendering of the experimental setup for measuring the free-swimming speed, with a close-up of the 3D-printed release mechanism. (**C**) Robot with tracking markers used for analyzing the undulatory kinematics. Note that the dimension annotations represent the unit scale bars for the 3D views (in A and B) and 2D views (in B and C), rather than the actual dimensions of the illustrated components.

(A_2/A_1), tailbeat amplitudes (θ_2), and frequencies (f_2). The CF must balance its size, tailbeat frequency, and stroke amplitude to generate sufficient thrust with least torque requirement. A higher-torque motor is required to drive the tailbeat mechanism, where a larger CF beats at a lower frequency and smaller amplitude (Fig. 4A). By contrast, a smaller fin with a larger stroke amplitude beating at a higher frequency should be considered for a smaller torque requirement. For example, to produce a thrust of 1.91 N, a CF beating at 10.0 Hz with a stroke amplitude of 20° requires a CF area five times larger than A_1, along with a driving torque of 2.01 N.m (#1, Fig. 4A). Note that A_1 is the caudal fin area of KUFish, equivalent to 19.14 cm^2. At a higher frequency (11.0 Hz) and a stroke amplitude of 40°, the required CF area is reduced to approximately $2.1A_1$, and the required torque decreases by 70.6% to 0.59 N.m (#2, Fig. 4A). For a stroke amplitude of 60° at 12.5 Hz, the same thrust production needs only a 76% smaller CF area (comparing $A_2/A_1 = 1.2$ and 5.0) and 86.1% less torque, with Q_2 being 0.28 N.m (compared with 2.01 N.m) (#3, Fig. 4A). For the tailbeat mechanism of KUFish-II, the DCX 22S-24V motor (Maxon, Switzerland, 66.0 g, 10,800 *RPM*, 0.12 N.m stall torque) was chosen. Then, the available torques by a combination of motor-transmission systems can be calculated for the three GRs of 10.0, 14.0, and 18.0, using Eq. (7). The three parallel dashed lines on the two vertical principal planes in Fig. 4A represent the available torque. The results revealed that when the CF beats with a stroke amplitude larger than approximately 30.0°, all three GRs satisfy the torque requirements. However, given the body length (L) and the assumed upper bound for the CF length ($L_c < 0.4L$), the tail stroke angle should be kept below 40° to ensure a reasonable peak-to-peak amplitude of less than $0.25L$ (see Step 2, Fig. 1). Furthermore, for efficient locomotion, swimming with a relatively small A_m can be achieved by decreasing the tail stroke angle. Accordingly, KUFish-II is designed with a *GR* of 18.0 and a stroke amplitude of 40° to create an A_m of approximately $0.25L$ (Step 2, Fig. 1).

Fig. 4B and C illustrate the comparison of the predicted frequency and required torque with the available values computed using Eq. (6)

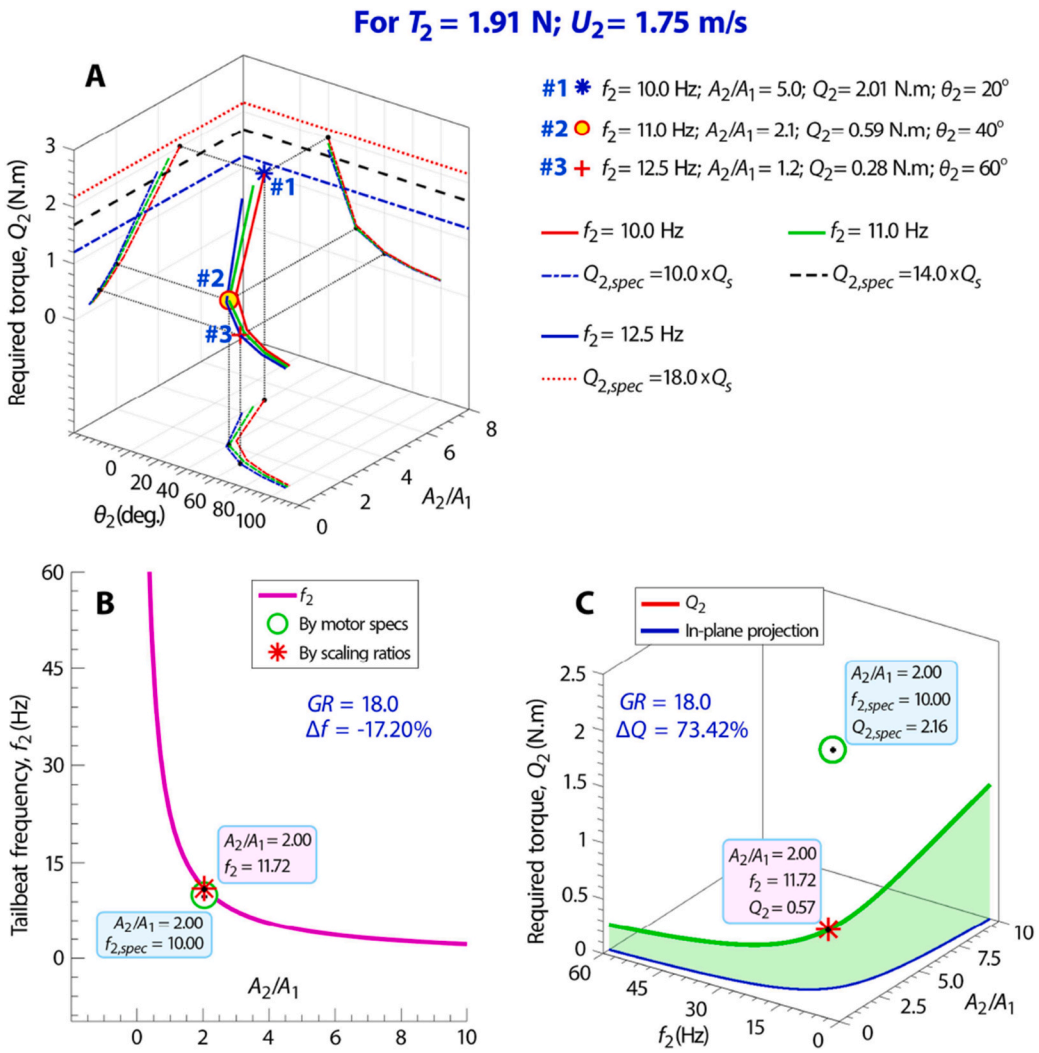


Fig. 4. System design. (**A**) Torque prediction from various tailbeat angles and CF areas at (10.0, 11.0, and 12.5) Hz. Three parallel dashed lines on two vertical planes are the available torques computed using Eq. (7). (**B-C**) Comparison of the (B) predicted and available frequencies using Eq. (4) and (6), respectively, and the (C) required and available torques computed using Eq. (5) and (7), respectively.

and (7), respectively. The selected A_2/A_1 of 2.0 is expected to result in beating at the predicted frequency of 11.7 Hz (Fig. 4B), which is almost identical to the target frequency of 11.5 Hz (Step 1, Fig. 1). As the selected *GR* was 18.0, the available $f_{2,spec}$ computed using Eq. (6) was 10.0 Hz, which was 17% lower than 11.7 Hz. However, this 17% higher frequency can be achieved by applying a higher voltage than the motor's nominal voltage. The required torque computed using Eq. (5) is approximately 73% smaller than the available $Q_{2,spec}$ (comparing 0.57 and 2.16 N.m, respectively) (Fig. 4C). This ensures that the selected motor with a *GR* of 18.0 can provide sufficient torque and drive the tailbeat mechanism at 11.7 Hz. Thus, it is expected that a CF area double that of the KUFish ($2A_1$) can beat at 11.7 Hz, produce a thrust of 1.91 N, and achieve the target speed of 1.75 m/s.

3.2. *Free-swimming undulatory kinematics*

Fig. 5A shows the trajectories of the free-swimming KUFish-II, which is powered by an 8-cell battery package (150 mAh, 3.7 V, Lithium-Polymer, Dualsky, China), supplying an input of approximately 30 V. With this voltage application, the undulatory frequency was measured as 11.9 Hz for the body and 12.2 Hz for the CF. Thus, these frequencies are considered practically equivalent. In addition, the trajectory was considered nearly straight even without control, with a deviation of only 2.1° from the swimming direction (blue dashed lines in Fig. 5A). Based on the tracking data in Fig. 5A and MATLAB analysis, Fig. 5B presents the pixel-based swim paths over three tailbeat cycles for the case with a maximum speed of 1.79 m/s. The corresponding swimming kinematics were processed and shown in Fig. 5C. Owing to the difficulty in maintaining swimming levels, the peaks of the measured undulatory angles increased. Therefore, the undulatory angle amplitudes were averaged over the first two cycles. The measured results show an amplitude of the peduncle angle of approximately 41.6° ($\Delta\theta_p$ in Fig. 5C), which is close to the designed tailbeat angle of 40.0°, thereby validating the design of the tailbeat mechanism. Due to tail flexibility, the tailbeat motion reaches a slightly larger amplitude of approximately 45.6° ($\Delta\theta_t$ in Fig. 5C). In contrast, the body undulates with a significantly smaller amplitude of about 15.4° ($\Delta\theta_b$ in Fig. 5C and D) and is completely out of phase with the tailbeat motion.

In addition to the 30-V case, the body undulation driven by a lower voltage battery package (26 V) was also examined (Fig. 5D). At 26 V, the body fairing undulates at 11.4 Hz, which is almost identical to the tailbeat frequency measured using the HSC. A comparison of the two voltage applications revealed that a 5.0% increase in frequency from 11.4 to 11.9 Hz reduces the body undulation angle by 6.0% from 16.4° to 15.4°, both of which slightly differ from the body undulatory angle of KUFish, which is approximately ±8°, at 10.64 Hz (Nguyen et al., 2022).

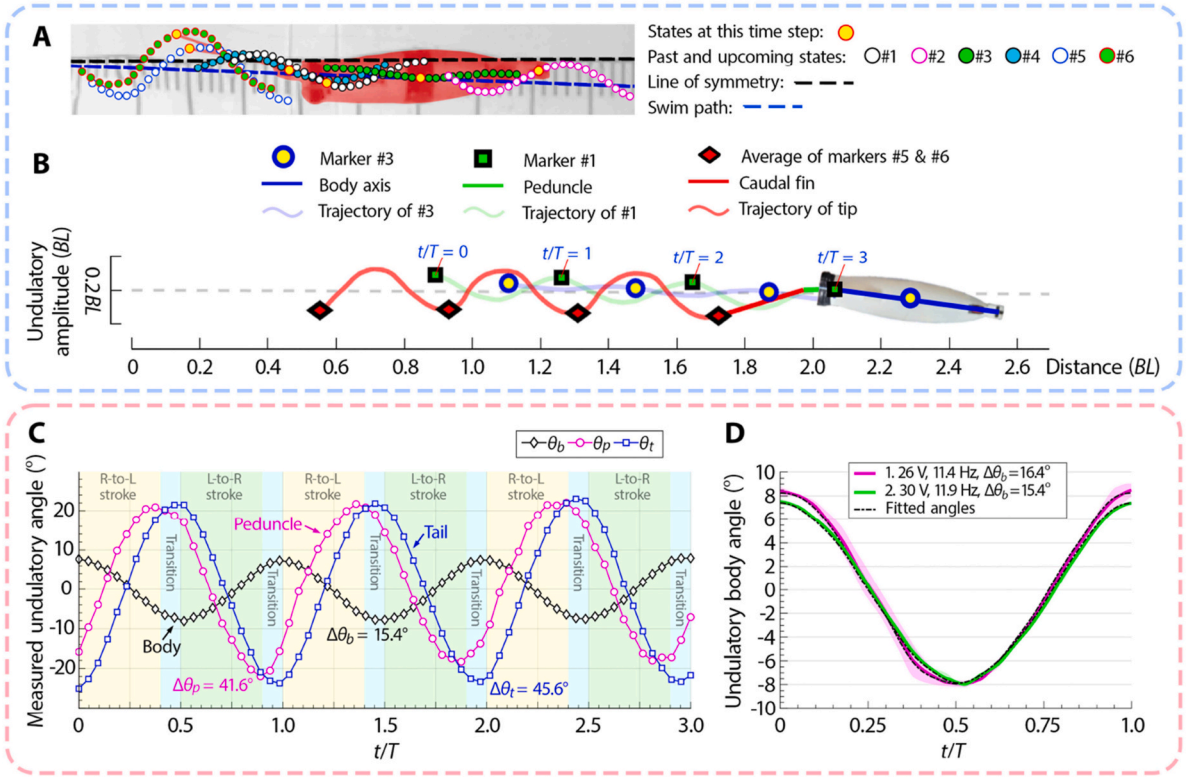


Fig. 5. Swimming kinematics. (**A**) Swimming trajectories of the six makers for the KUFish-II. (**B**) Reconstruction of two-dimensional (2D) pixel-based swim paths using MATLAB, based on the tracking data shown in (A). The horizontal grey dashed line represents the 1-m symmetry line of the pool, equivalent to 610 pixels or 2.7 BL. (**C**) Measured periodic undulatory angles for the body (θ_b), peduncle (θ_p), and CF (θ_t). (**D**) Measured and fitted body undulatory angles from two swimming tests powered by the 26-V and 30-V battery packages (see Appendix E for further details of the Fourier fitting functions). The shaded zones represent the standard deviation. Marker numbering is detailed in Fig. 3C.

Given the modified body shape and the slight changes in the frequency and undulation angle relative to KUFish, the initially assumed body drag coefficient of KUFish-II was reassessed using CFD analysis. Note that the CFD analysis focused solely on the body, excluding the caudal fin, as the drag defined in this study applies specifically to the body, while the tail generates thrust. Thus, the longitudinal force coefficients (C_{Fx}) by the body were computed for the two cases shown in Fig. 5D, which were used to determine the body drag coefficient (C_D). In Appendix F, we explain how the body undulatory motions corresponding to the 26-V and 30-V real-swimming cases were implemented in ANSYS Fluent (ANSYS Inc., Canonsburg, PA, USA).

3.3. *Measured frequency, thrust, power, and free-swimming speed*

Fig. 6A shows the measured thrust by the $2A_1$ CF at frequencies ranging from 8.0 to slightly over 12.0 Hz. To validate the reliability of the power measurements, we compared the independent thrust signals recorded during the thrust and power measurements (thrust shown in Fig. 6A and B, respectively; experimental setup shown in Fig. 3A). The comparison revealed a minor percentage difference in thrust between the two datasets. For example, in the thrust measurement (Fig. 6A), the interpolated thrust at the predicted tailbeat frequency of 11.7 Hz is 1.99 N. Meanwhile, in the power measurement, the interpolated thrust at the same frequency is 1.90 N (Fig. 6B). These measured values are within a 4% agreement with the prediction (1.91 N) by Eq. (2) at 11.7 Hz. This confirms the accuracy of the thrust measurement, thereby validating the power measurements for KUFish-II. The thrust values reported in this study were obtained from the thrust measurements in Fig. 6A.

The measured swimming speed and tailbeat frequency of the KUFish-II are summarized in Fig. 6C. The interpolated thrusts at two frequencies of 11.4 and 12.2 Hz, measured during free swimming at 26 V and 30 V, respectively (Fig. 6C), are shown in Fig. 6A as 1.92 N and 2.07 N. Despite meeting the thrust requirement of 1.91 N, KUFish-II with a 26 V input swam at 11.4 Hz, which is 3% lower than the predicted tailbeat frequency of 11.7 Hz ($T = 1.92$ N at 11.4 Hz; Fig. 6A). Therefore, its maximum swimming speed is only 1.56 m/s, resulting in an average of 1.52 ± 0.04 m/s (Fig. 6C). To slightly increase the tailbeat frequency, one more battery was added, forming a 30 V supply. This enabled the KUFish-II to increase the frequency from 11.4 to 12.2 Hz, produce 7.8% more thrust (2.07 N compared to 1.92 N, Fig. 6A), reach a maximum speed of 1.79 m/s, and maintain an average speed of 1.67 ± 0.04 m/s (Fig. 6C). Therefore, the robot with the $2A_1$ CF under 30 V meets the goal while consuming a power of 31.7 W (Fig. 6B). Additionally, the pixel-based measurement showed that the A_m was approximately $0.13L$ (Fig. 5A and B). For validation, the calculated A_m using the measured maximum undulatory angles of the body, peduncle, and CF in Fig. 5C was also within $(0.13-0.14)L$ (see Fig. G1 and Appendix G for details). This revealed that KUFish-II swam at a *St* of 0.28, which is relatively smaller than those of the other robotic platforms (Fig. 6D). This value also fell within the optimal range observed in nature (yellow shaded area in Fig. 6D). Thus, KUFish-II can be considered an efficient swimmer (Clapham and Hu, 2014; Iguchi et al., 2024; Wen et al., 2013; White et al., 2021; Zhong et al., 2017; Zhu et al., 2019).

Although the swimming speed primarily depends on the thrust produced by the CF, minimizing body drag is also a critical factor in enhancing the swimming performance (Webb, 1988). Therefore, estimating body drag during undulatory swimming is essential to complete the design process. Using CFD analysis based on the measured body undulatory kinematics, the body drag coefficient of KUFish-II was numerically determined. In the CFD analysis, we applied the same meshing technology used to describe the dynamic mesh motions of the

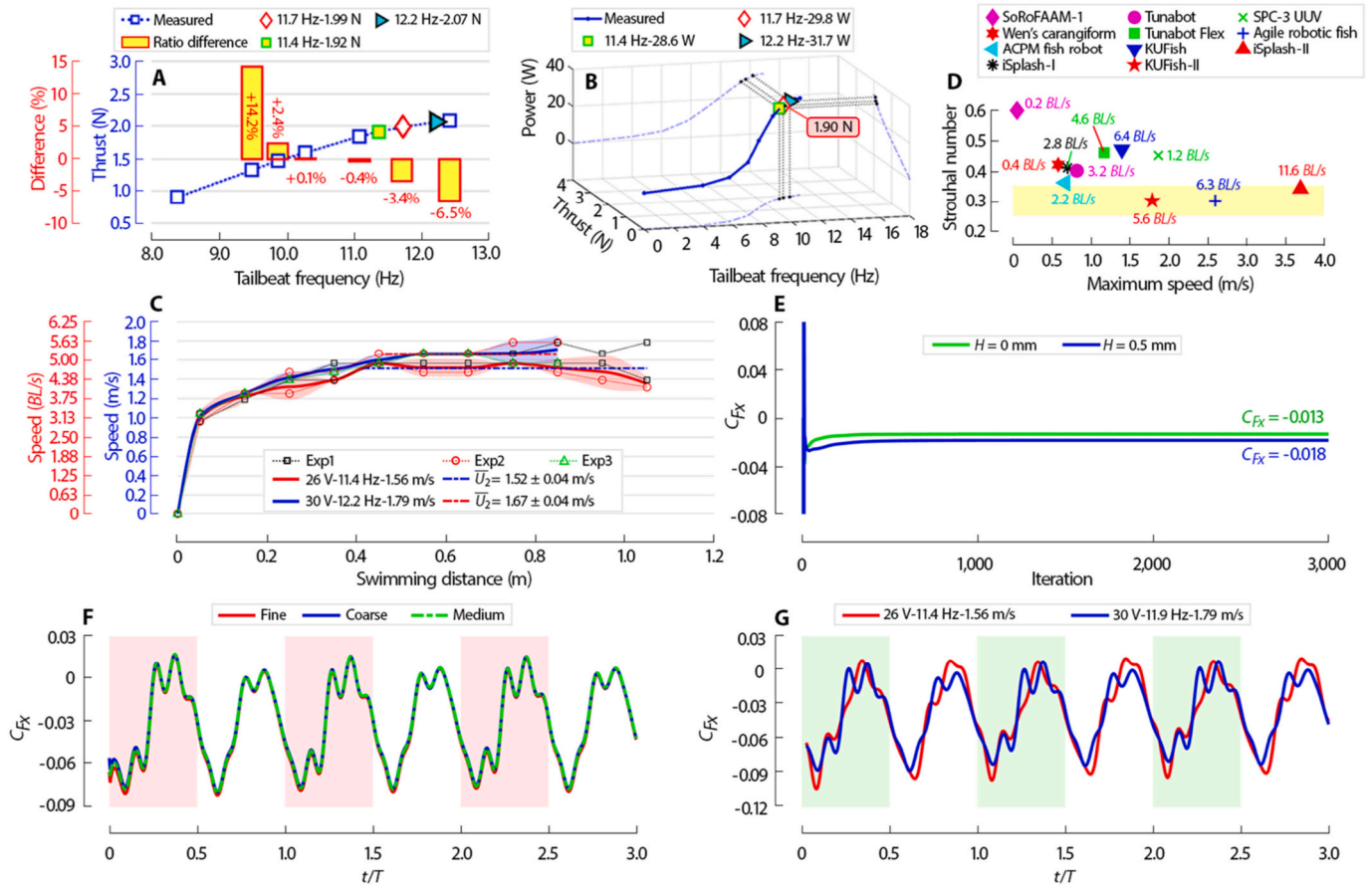


Fig. 6. Experimental and CFD results. (**A-D**) Experimental results of (A) cycle-average thrust by the $2A_1$ CF across varying tailbeat frequencies (shaded areas show standard deviation from three trials, 95.0% confidence level), (B) averaged measured power consumption versus tailbeat frequency, alongside corresponding thrust production, (C) swimming speeds of 26-V and 30-V KUFish-II (shaded regions indicate standard deviation from two trials for 26-V cases, and three trials for 30-V cases, 95.0% confidence level), and (D) comparison of St and maximum speeds for KUFish-II and other robots with the shaded region showing $0.25 < St < 0.35$. See Movies S2 to S8 for tethered and free-swimming tests. (**E-G**) CFD results: (E) Body drag coefficient during steady, static swimming. (F) Mesh convergence study of C_{Fx} for zero-roughness model undulating at 11.9 Hz (30 V) across three mesh types, using the corresponding measured undulatory kinematics in Fig. 5D. (G) Time courses of C_{Fx} for two models at the body undulatory frequencies of 11.4 Hz (26 V) and 11.9 Hz (30 V), based on their respective measured kinematics shown in Fig. 5D. Note: roughness height is 0 mm in (F) and 0.5 mm in (G). See Appendix F and Fig. F1 for more details of the CFD modeling and analysis.

flapping wing and tailbeat motion (Nguyen et al., 2021a, 2021b, 2024; Nguyen and Park, 2023). Here, Fig. 6E–G and Table 2 summarize the CFD results on the effects of undulatory frequency on longitudinal force production along the body's *x*-axis. A negative cycle-average C_{Fx} indicates the body drag, whereas a positive value denotes the thrust. Therefore, the drag coefficient is equivalent to the absolute value of C_{Fx} in Table 2.

A mesh convergence study using the zero-roughness model across three different mesh types is shown in Fig. 6F, demonstrating that the results obtained with the medium mesh converged at the third cycle. Table 2 compares the cycle-average C_{Fx} by 26-V and 30-V inputs, corresponding to the body undulatory frequencies of 11.4 and 11.9 Hz, respectively. The results showed that a 4.4% increase in undulatory frequency led to a 6.1% decrease in the body's undulatory angle, which in turn resulted in a 5.4% reduction in the total C_{Fx}. Interpretations of the relationship among swimming speed, body undulatory angle, and body drag coefficient are detailed in Section 4. Notably, the C_{Fx} value of −0.035 ($H = 0.5$ mm, Table 2) obtained by 30-V undulatory swimming was double that of −0.018 computed from the static swimming simulation (see $H = 0.5$ mm, Fig. 6E).

Table 2
Cycle-average body drag coefficient.

Vol_2 (V)	f_b (Hz)	$\Delta\theta_b$ (°)	C_{Fx}*	C_{Fx}**	C_D***
26	11.4	16.4	N/A	−0.037	0.037
30	11.9 (+4.4%)	15.4 (−6.1%)	−0.029	−0.035 (−5.4%)	0.035

*The simulated C_{Fx} is obtained at a roughness height of $H = 0$ mm.
**The simulated C_{Fx} is obtained at a roughness height of $H = 0.5$ mm.
***The drag coefficient is defined as the absolute value of C_{Fx}.

4. Discussion

The interpretation of physical mechanisms correlated with the design parameters (relations among thrust, torque, tail stroke angle, fin area, and tailbeat frequency) is actually challenging. Therefore, to establish the scaling laws, we have chosen to rely on the parametric studies based on CFD simulation and the added mass method (Pham, 2022) for a simplified fin shape to achieve Eq. (3)–(5). While the effect of fin area and tailbeat frequency can be also confirmed by equations used to derive the added mass method, the effect of tailbeat amplitude was identified through the presented parametric studies in Appendix D. Therefore, even when *AR* of tail fin becomes significantly high, the exponents for fin area and frequency in Eq. (3)–(5) would not change. However, the exponent for tailbeat stroke amplitude can alter. Therefore, for caudal fins with a significantly large *AR*, we need to perform another set of CFD calculations and extract the exponent for the stroke amplitude.

The proposed scaling method is applicable for scalable robotic fish

with body shapes similar to those of the reference models, assuming an identical C_D. A smaller difference in C_D leads to more accurate thrust prediction using the 1D EoM, making the scaling method more reliable. However, even if the body shape is different, the required thrust for maximum speed can still be predicted by using the 1D EoM once the body area and C_D are prescribed. Subsequently, the same scaling laws in Eq. (3)–(5) can be applied to determine the required tailbeat frequency and torque for the target speed. However, the shapes of caudal fins need to be identical. The measured frequency, thrust, and swimming speed were in good agreement with the predicted values. Therefore, the current scaling-based design can provide the key specifications for motor selection. With a properly selected motor and *GR*, the fabricated tailbeat mechanism produced the required thrust by beating the CF with the predicted area and frequency. Thus, the scaling-based design proposed in this paper may provide a systematic design methodology for robotic fish. The swimming speed predicted by the three physical parameters is a good estimation for the target speed. At the tailbeat frequency of 12.2 Hz, the KUFish-II swam mostly straight even without feedback control, generated a thrust of approximately 2.1 N, and reached a maximum speed of 1.79 m/s. In addition, the robot achieved a maximum *Fr* of 1.01 and a Reynolds number on the order of 10^5 (see Section F.1), consistent with high-burst swimming regime at 1.79 m/s (5.6 BL/s) (Bandyopadhyay, 2005). The swimming speed prediction using three physical parameters indicates that the body length must be longer to swim faster than 1.79 m/s.

With the tethered measurement, we compared the scaling ratios of the thrust with those of the square of the tailbeat frequency to investigate the accuracy of the thrust measurement using Eq. (3) for the same θ and A. The differences between the ratios of (T_i/T_{i-1}) and $(f_i/f_{i-1})^2$ measured at two consecutive frequencies are indicated by yellow columns (Fig. 6A). For the measured frequencies of 9.9, 10.3, 11.1 Hz, and 11.7 Hz, the ratio differences are less than 3.4%. This indicates that the thrusts at these frequencies were accurately measured. However, the differences are 14.2% for frequencies between 8.4 and 9.5 Hz and −6.5% for frequencies between 11.7 and 12.4 Hz. Thus, for a given fin stiffness, there exists a frequency range at which thrust generation is close to the predicted thrust (1.91 N) at the estimated frequency (11.7 Hz). For the same fin areas, the thrust ratio over two consecutive tests should align with the scaling thrust: $(T_i/T_{i-1}) \approx (f_i/f_{i-1})^2 \times (\theta_i/\theta_{i-1})^{2.15}$. In the thrust measurement, it was observed that the difference in stroke angles appeared negligible over two trials $(\theta_i/\theta_{i-1})^{2.15} \approx 1$, at around the predicted frequency, resulting in a simplified thrust ratio that quadratically scales with only the frequency ratio: $T_i/T_{i-1} \approx (f_i/f_{i-1})^2$. As shown in Fig. 6A, the ratio differences are nearly zero at around 11 Hz. However, at frequencies deviating from 11 Hz, the difference grows. During the experiments, we observed that the body's undulatory angle increased at lower frequencies (comparing 16.4° at 11.4 Hz and 15.4° at 11.9 Hz; Fig. 5D), which lead to larger tail stroke amplitudes and consequently higher thrust than predicted by the frequency-based scaling (denoted by positive differences; Fig. 6A). Meanwhile, at high frequencies, the fin deformed in another mode (larger deformation near the peduncle), leading to smaller stroke amplitudes at the fin tip and thus producing lower thrust than predicted by the frequency-based scaling (denoted by negative differences; Fig. 6A). Therefore, the thrust may not be purely proportional to the square of the frequency. In summary, the relatively larger discrepancies at frequencies deviating from 11 Hz in Fig. 6A can be attributed to the omission of the actual tail stroke angle at low frequencies and unmolded tail flexibility at high frequencies in this frequency-based scaling comparison, which assumes identical tailbeat amplitudes at all frequencies. Thus, although the scaling-based thrust prediction is reliable near the design frequency, its accuracy cannot be guaranteed at frequencies significantly above or below the design frequency. This clearly highlights the influence of the actual tail stroke angle and the unmodeled fin stiffness in accounting for the ratio difference observed in the comparison. Indeed, according to a review (Triantafyllou et al., 2005), propulsive efficiency and fin stiffness were significantly correlated. A significant increase in propulsive efficiency was theoretically (Bose, 1995) and experimentally (Prempraneerach et al., 2003) confirmed for a flexible foil relative to a stiff foil, whereas its produced thrust remained almost unchanged (Prempraneerach et al., 2003). This indicates that the influence of fin stiffness should be systematically considered in the scaling laws (Triantafyllou et al., 2005). Based on our experience, we suggest that the caudal fin stiffness should be properly chosen such that the caudal fin deforms like the first beam bending mode keeping $0.1L < A_m < 0.3L$ at the desired tailbeat frequency.

The measured swimming kinematics indicated that KUFish-II swam efficiently with a *St* of approximately 0.28, falling within the optimal range of 0.25 to 0.35 for natural species. Similar conclusions were drawn from studies on oscillating airfoils (Anderson et al., 1998; Lauder and Tytell, 2005; Read et al., 2003; Triantafyllou et al., 1991; Wen et al., 2013). In fact, the propulsive efficiency dropped significantly within the range of $0.35 < St < 1.025$ after peaking at $St = 0.3125$ (Wen et al., 2013). Similarly, the authors of (Read et al., 2003) tested *St* outside the optimal range to compare the thrust coefficient and propulsive efficiency of underwater oscillating foils. The results showed a slight decrease in the thrust coefficient (0.62 vs. 0.58) and a 20% drop in efficiency (60% vs. 40%) as *St* increased from 0.4 to 0.6, indicating that both were caused by the transition from thrust to drag. By comparison, the original KUFish swam at a speed of 1.40 m/s with a *St* of 0.45, whereas the KUFish-II achieved a faster speed of 1.79 m/s with a 37.8% lower *St* of 0.28. This shows that the swimming speed increase from 1.40 to 1.79 m/s is associated with a decrease in *St* from 0.45 to 0.28. A similar trend in the relationship between swimming speed and *St* was also observed in real fish (Lauder and Tytell, 2005) and further validated using the Tunabot robotic platform (Zhu et al., 2019). Although a higher *St* results in an increase in thrust, it may produce significant drag, causing a slowdown in swimming speed (Read et al., 2003; Wen et al., 2013).

Regarding energy expenditure of swimming, at 12.2 Hz, the KUFish-II consumed 31.7 W to reach 1.79 m/s (5.6 BL/s), which corresponds to a cost of transport (*COT*) equal to 17.7 J/m, defined as the ratio of power to swimming speed (Lauder et al., 2012; Tangorra et al., 2011; Zhu et al., 2019). According to (Saadat et al., 2017), by swimming with $St = 0.28$ and $A_m = 0.13L$, KUFish-II could attain efficient locomotion with low power consumption and, consequently, a low *COT*, calculated as the ratio of power consumption to maximum speed. More interestingly, this low *COT* of KUFish-II is only 17% of that of the 1.6-m-long SPC-3 UUV (17.7 J/m vs. 101.6 J/m), which swam at a similar speed of 1.87 m/s (1.2 BL/s) but consumed approximately 190.0 W (Wang et al., 2010). Comparing with the agile robotic fish that consumed 170 W and exhibited a speed of 2.6 m/s (or a similar speed of 6.3 BL/s) (Iguchi et al., 2024), the *COT* of KUFish-II swimming at 5.6 BL/s is 73% less, resulting in its *COT* of 65.4 J/m. Notably, although the agile robotic fish also achieved low energy swimming due to $St = 0.3$ and $A_m = 0.19L$ (see Table 1), the robot consumed more power (170 W vs. 31.7 W) due to its longer body for swimming at a comparable speed (6.3 BL/s). This means that the KUFish-II is among the few high-speed robotic swimmers capable of achieving energy-efficient swimming. Note that we evaluated the *COT* only at the measured maximum speed, as our primary interest is achieving efficient high-speed swimming.

To validate the initial assumption on C_D of the KUFish-II and evaluate the relationship among body undulation, body drag, and swimming speed, a CFD simulation was conducted. The CFD analysis revealed that the rightward body undulation shifts the stagnation point and high-pressure region at the nose to the right (highlighted in red; see Step 6, Fig. 1). Simultaneously, the low-pressure region (blue) on the left side moves anteriorly, creating a pressure difference across the nose. This subsequently causes water to flow from the high-pressure to low-pressure side, generating a leftward force owing to the leading-edge

suction from the airfoil-like shape of the anterior body (Step 6, Fig. 1). See Movie S9 for details. This phenomenon aligns with the findings of a previous study (Lucas et al., 2020) on thrust generation in the negative pressure region of the anterior airfoil-like body. The resulting force also helps reduce the undulatory body angle by generating a counterclockwise restoring moment about the center of buoyancy (CB). As the robot swims faster, a greater leftward force is produced during the rightward undulation owing to the increased inflow speed relative to the body (Step 6, Fig. 1). A similar mechanism occurs during leftward undulation, when a rightward force and an associated clockwise restoring moment are produced. This explains why the KUFish-II exhibited smaller body undulation at higher swimming speeds (15.4° at 30 V, 1.79 m/s) than at slower speeds (16.4° at 26 V, 1.56 m/s) (Table 2). The reduced undulatory angle subsequently decreased the frontal area, thereby lowering the body drag (comparing 0.037 and 0.035, Table 2). Compared with the C_D of the KUFish robot, the C_D of KUFish-II was about 9% higher (0.035 vs. 0.032), even though they were assumed to be identical in Eq. (2). The 9% difference in C_D may be caused by slight difference between the initial body shape used in the scaling design and actual body shape used for fabrication, highlighting the importance of maintaining body-shape similarity.

Under static, steady swimming (without undulatory motions of body and caudal fin), the C_D of KUFish was experimentally measured and compared with the CFD-computed value in our previous study (Nguyen et al., 2022). Based on the measured surface roughness of 0.16 mm, the CFD-computed C_D of KUFish was 13% lower than the measured value (0.0134 vs. 0.0160). For conservative prediction in this study, CFD for KUFish-II used an assumed roughness of 0.5 mm, yielding a static C_D of 0.018 (Fig. 6E), which is 13% larger than the measured value of KUFish. Therefore, the 13% larger C_D results from the slightly modified body shape and higher roughness. In addition, under undulatory motion, C_D increased to 0.035, which is about twice the static value and consistent with the trend observed in KUFish. This provides indirect validation of the C_D used for KUFish-II.

In evaluation of swimming performance by the 1D EoM in Eq. (1) and the measured thrust (2.07 N), the C_D of KUFish-II, whose actual wetted area is 0.0262 m^2, was estimated to be 0.05 in order to reach the maximum speed of 1.79 m/s. For swimming distance, the actual body mass of approximately 0.25 kg with an added mass of about 20% more was estimated to reach 99% of the maximum speed after traveling 0.8 m. For both predicted swimming distance and swimming speed, the EoM provided a good estimation of the measured values in the free swimming test (the robot reached 1.79 m/s at around 0.75 m, see Fig. 6C). It means that the predicted C_D of 0.05 is close to the actual value in free swimming. The CFD-based C_D is lower than the 1D EoM prediction (0.035 vs. 0.05), indicating existence of slight pitching motion during practical swimming tests, explaining the underestimation of body drag by CFD simulation. However, the numerical method clarified the relationship among body drag, body undulatory angle, and free-swimming speed. This study serves a reference for numerical thrust estimation at the maximum speed ($T = D$) in larger-scale models when direct force measurements are difficult.

As our focus is on both high-speed and efficient swimming, the proposed design approach targets swimming with $Fr = 1$ (for high speed). Since only iSplash-II and Agile Robotic Fish have achieved swimming with $Fr > 1$ (Table 1), assuming $Fr = 1$ for fast- or burst-swimming regime is a conservative and reasonable choice. High swimming efficiency is then pursued by placing St between 0.25 and 0.35 with $0.1L < A_m < 0.3L$. For general use of the proposed approach to design robotic fish for relatively slow but efficient cruising, we may first choose a target speed that makes $Fr < 1$ for a given body and determine another speed and frequency to place St between 0.25 and 0.35 with $0.1L < A_m < 0.3L$, such that the speed is close to that predicted by the chosen $Fr < 1$. The reduced frequency (σ) is used to confirm the predetermined speed and the frequency by checking if σ ranges from 10 to 20, according to Table 1. Therefore, the proposed method can be used for designing robotic fish with $Fr < 1$.

5. Conclusion

In this work, a scaling-based design approach was proposed and demonstrated to be effective for developing an energy-efficient robotic fish capable of achieving a target swimming speed. The target swimming speed of the 0.32-m-long KUFish-II was reasonably estimated to be 1.75 m/s using Froude number, Strouhal number, and reduced frequency. Upon fabrication and testing, the robot was able to produce the required thrust and tailbeat frequency to achieve the target speed. In free-swimming tests, the robot successfully achieved a maximum speed of 1.79 m/s, closely matching the target. The tailbeat stroke angle of 40° was systematically designed to achieve the efficient swimming, leading to a tail peak-to-peak amplitude of about 13% of the body length and a Strouhal number of 0.28, both within the optimal ranges. This indicates that the KUFish-II is among the few robotic swimmers that can swim efficiently while consuming minimal energy. These results show the effectiveness and applicability of the proposed scaling-based design approach to develop fast-swimming and scalable robotic fish with minimal energy consumption.

CRediT authorship contribution statement

Khanh Nguyen: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hoon Cheol Park:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Figure

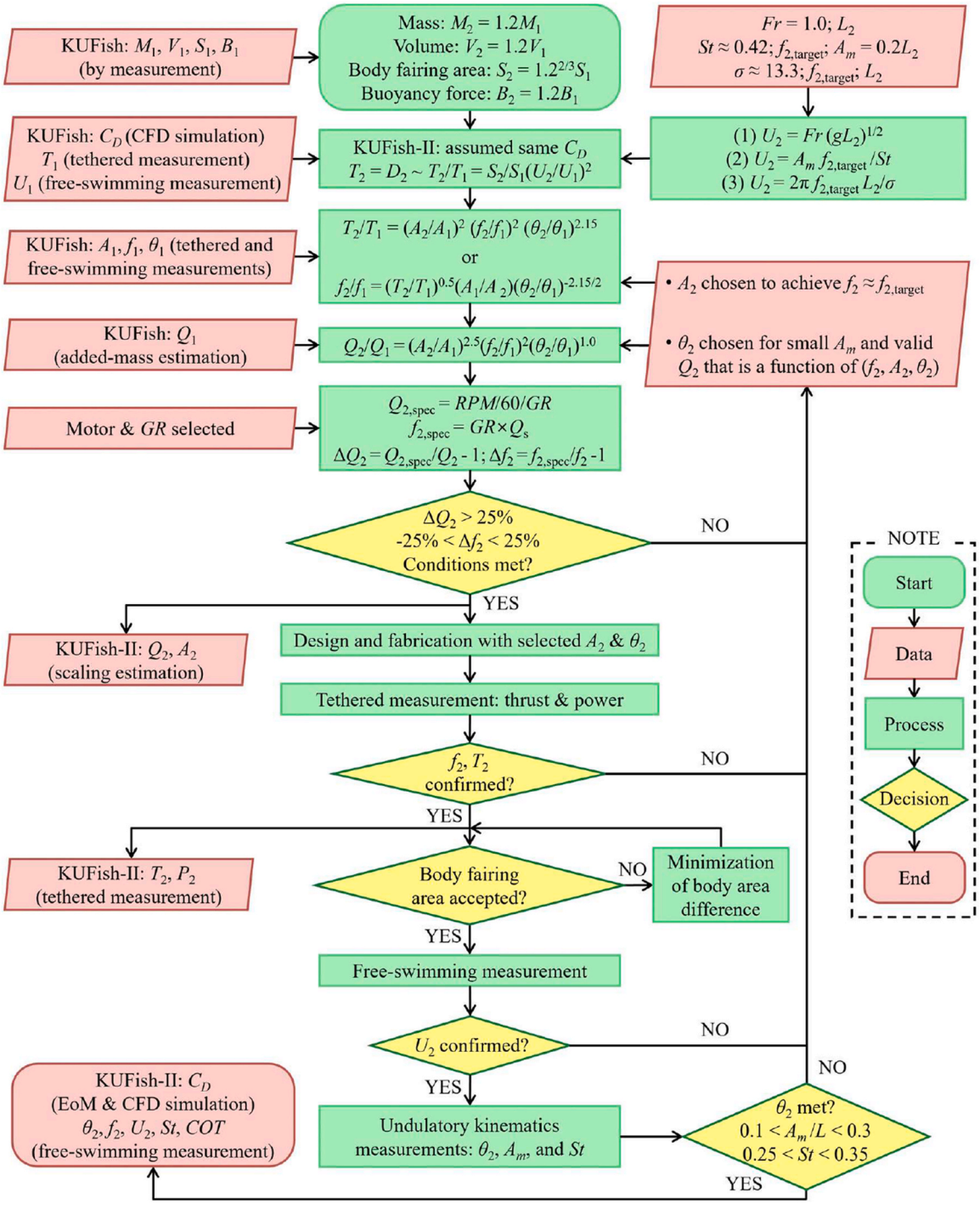


Fig. A1. Flow chart for the development of KUFish-II. A 25% difference between the motor's available torque and frequency and the corresponding values of caudal fin was allowed in the motor evaluation step based on practical experiences. For frequency, a negative deviation can be compensated by applying a higher voltage. However, for safety and prolonged duration, a positive deviation is recommended.

Appendix B. Table

Table B1
Comparison between the predicted and actual (measured) values of three physical parameters and their corresponding swimming speeds

Parameters	Units	Predicted	Actual	Difference
Total body length	m	0.32	0.32	
Peak-to-peak amplitude/total length	%	20.00	13.00	−35.0%
Target tailbeat frequency	Hz	11.50	12.20	6.1%
Froude number, *Fr*	-	1.00	1.01	1.0%
Strouhal number, *St*	-	0.42	0.28	−32.6%
Reduced frequency, σ	-	13.30	13.70	3.0%
Swimming speed estimated by *Fr*	m/s	1.77	1.79	1.0%
Swimming speed estimated by *St*	m/s	1.75	1.79	2.3%
Swimming speed estimated by σ	m/s	1.74	1.79	3.0%
Average estimated swimming speed	m/s	1.75	1.79	2.1%

Table B2
Actual parameters of KUFish and estimated ones of KUFish-II

Parameters	Symbols	KUFish*	KUFish-II	Units
Body length	L	0.22	0.32	m
Length of CF	L_C	0.06 ($= 0.27L$)	$< 0.4L = 0.12$	m
Total mass	M	$M_1 = 0.116$	$M_2 = 1.2 \times M_1 = 0.139$	kg
Motor mass	M_m	$M_{m1} = 0.025$	$M_{m2} = 0.066$	kg
Body volume	V	$V_1 = 0.000116$	$V_2 = 1.2 \times V_1 = 0.000139$	m^3
Body wetted area	S	$S_1 = 0.0211$	$S_2 = (1.2)^{2/3} \times S_1 = 0.0238$	m^2
Swimming speed	U	1.40	1.75	m/s
Tailbeat frequency	f	11.4	11.7	Hz
Thrust	T	1.08	1.91	N
Area of CF	A	$A_1 = 1.914 \times 10^{-3}$	$A_2 = 2A_1$	m^2
Fin span	b	0.07	0.1	m
Aspect ratio	AR	2.56	2.56	
Required torque	Q	0.144	0.57	N.m
Reduction gear ratio	GR	14.4	18.0	

* The selected reference parameters are detailed in (Pham et al., 2023)

Appendix C. Effect of target frequency selection on swimming performance predictions

For a trial selection of 5.0 Hz, while maintaining the same L and A_m, the parameters of $Fr = 1.0$, $St = 0.18$, and $\sigma = 5.8$, are selected, which result in the possible swimming speeds of 1.77, 1.75, and 1.73 m/s, respectively. In this case, the average swimming speed is again 1.75 m/s. Although these computed swimming speeds are close to each other with small differences, both the selections of $\sigma = 5.8$ and $St = 0.18$ fall beyond the typical ranges reported for swimming robots. This indicates a feasible limitation of such low-frequency designs. Similarly, at a higher trial frequency of 20.0 Hz and with the same L and A_m, the set of ($Fr = 1.0$, $St = 0.74$, and $\sigma = 23.0$) is selected to achieve the same target speed of 1.75 m/s. However, $St = 0.74$ is over its optimal range, and $\sigma = 23.0$ is significantly high, indicating an unfeasibly high-frequency design under the current assumptions. Therefore, to predict the possible swimming speed, it is important to choose design parameters, such as the body length tailbeat frequency, and peak-to-peak amplitude, such that three parameters *Fr*, *St*, and σ fall within feasible ranges at the preliminary design stage.

Appendix D. Updates for the scaling-based equations of predicted thrust and torque

Table D1 summarizes the data used to derive Eq. (3). These data, which were obtained in (Nguyen et al., 2023), were extracted from a series of CFD analyses using ANSYS Fluent. More details on the CFD modeling and simulation can be found in (Nguyen et al., 2023). More specifically, using the relationships between $T \sim \left(A^2; f^2\right)$ and $Q \sim \left(A^{2.5}; f^2\right)$, as presented in (Nguyen et al., 2023), Eq. (D1) and (D2) describe the ratios of thrust (T_2/T_1) and torque (Q_2/Q_1) for the caudal fin areas (A_2/A_1) and tailbeat frequencies (f_2/f_1), where subscripts "1" and "2" denote the two different cases being investigated.

$$\frac{T_2}{T_1} = \left(\frac{A_2}{A_1}\right)^2 \times \left(\frac{f_2}{f_1}\right)^2. \tag{D1}$$

$$\frac{Q_2}{Q_1} = \left(\frac{A_2}{A_1}\right)^{2.5} \times \left(\frac{f_2}{f_1}\right)^2. \tag{D2}$$

Table D1
Cycle-average thrust by the CFD simulations and exponent "x" values

Tailbeat frequency (Hz)		10 Hz			20 Hz			30 Hz		
Tailbeat stroke amplitude (°)		20°	50°	80°	20°	50°	80°	20°	50°	80°
Thrust by different CF area ratios (N)	1.0	0.1	0.9	2.3	0.4	3.6	9.0	1.0	8.0	20.3
	2.25	0.5	4.5	11.3	2.2	18.0	45.3	4.8	40.5	101.9
	4.0	1.8	14.3	36.4	7.1	57.3	145.7	15.9	128.9	327.9
Stroke amplitude ratios			50°/20°	80°/50°		50°/20°	80°/50°		50°/20°	80°/50°
Exponent "x" by different CF area ratios	1.0		2.40	2.00		2.40	1.95		2.27	1.98
	2.25		2.40	1.96		2.29	1.96		2.33	1.96
	4.0		2.26	1.99		2.28	1.99		2.28	1.99
Average "x"			2.15							

However, the relationship between thrust and torque with respect to stroke amplitude has not yet been established in (Nguyen et al., 2023). For the same tail area ($A_2 = A_1$) and frequency ($f_2 = f_1$), the ratios of thrust, and torque are assumed to be proportional to the xth and y^{th} powers of the change in stroke amplitude, respectively, as follows:

$$\frac{T_2}{T_1} = \left(\frac{\theta_2}{\theta_1}\right)^x \text{ and } \frac{Q_2}{Q_1} = \left(\frac{\theta_2}{\theta_1}\right)^y. \tag{D3}$$

Then, the unknown exponents "x" and "y" are computed using Eq. (D4).

$$x = \log\left(\frac{T_2}{T_1}\right) \Big/ \log\left(\frac{\theta_2}{\theta_1}\right) \text{ and } y = \log\left(\frac{Q_2}{Q_1}\right) \Big/ \log\left(\frac{\theta_2}{\theta_1}\right). \tag{D4}$$

To determine the unknown exponent "x", we use the CFD results in Table D1. Specifically, at different tailbeat frequencies, we calculated the ratios of the thrust produced by different stroke amplitudes, along with their corresponding stroke amplitude ratios. The exponent "x" was computed using the logarithmic relationship in Eq. (D4). Table D1 summarizes the computed "x" values, with an average of 2.15. Then, by accounting for the stroke amplitude, the relationship among the thrust, CF area, and tailbeat frequency in Eq. (D1) was then updated as follows:

$$\frac{T_2}{T_1} = \left(\frac{A_2}{A_1}\right)^2 \times \left(\frac{f_2}{f_1}\right)^2 \times \left(\frac{\theta_2}{\theta_1}\right)^{2.15}. \tag{D5}$$

Table D2
Peak torque values for the CF of KUFish-II and the corresponding "y" values across three tailbeat frequencies and amplitudes

Tailbeat frequency (Hz)	10 Hz			20 Hz			30 Hz		
Tailbeat stroke amplitude (°)	20°	50°	80°	20°	50°	80°	20°	50°	80°
Peak torque (N.m)	0.4	1.0	1.6	1.6	4.0	6.5	3.6	9.1	14.5
Exponent "y"	1.00	1.00		1.00	1.00		1.00	1.00	
Average "y"	1.00								

To determine the unknown exponent "y" in Eq. (D4), we employed the added mass method described in (Pham, 2022) to compute the peak torques generated by the $2A_1$ CF of the KUFish-II for three stroke amplitudes of (20°, 50°, and 80°) and the tailbeat frequencies of (10, 20, and 30) Hz (see Table D2). By calculating the torque ratio corresponding to the change in stroke amplitude, the exponent "y" is shown in Table D2, with an average of 1.0.

Finally, by incorporating the stroke amplitude, the relationship among the torques, tail fin areas, and tailbeat frequencies in Eq. (D2) is then updated as follows:

$$\frac{Q_2}{Q_1} = \left(\frac{A_2}{A_1}\right)^{2.5} \times \left(\frac{f_2}{f_1}\right)^{2.0} \times \left(\frac{\theta_2}{\theta_1}\right). \tag{D6}$$

Appendix E. Fourier functions used to fit measured undulatory kinematics

The measured undulatory kinematics in Fig. 5D were fitted using the Fourier functions as follows:

$$f(\hat{t}) = a_0 + \sum_{k=1}^{k=9} [a_k \times cos(2k\pi f(\hat{t} + \hat{t}_s)) + b_k \times sin(2k\pi f(\hat{t} + \hat{t}_s))], \tag{E1}$$

where the fitting coefficients a_0, a_k, and b_k were incorporated into a user-defined function (UDF) in ANSYS Fluent to implement the prescribed ki-

nematics. Here, $\hat{t}$ shows the nondimensional time (t/T), and $\hat{t}_s = 0.25$ denotes the phase shift at which the body axis is close to the swimming axis, ensuring mesh symmetry for each half-stroke.

Appendix F. CFD modeling and analysis

F.1. CFD modeling

In Fig. F1, the body was positioned at the center of the longitudinal plane of symmetry, with its origin aligned to the tailbeat rotation axis (Fig. 2B). Three parameters including l_1, l_2, and l_3 (see Figs. F1C-F), were used to create the mesh, where l_1 is the maximum edge length of the triangular elements on the body surface, l_2 is the maximum edge length of the tetrahedral cells within the HDR, and l_3 is the maximum edge length of the triangular elements at the boundary of the refinement box. Similar to our previous CFD studies on periodic motion (Nguyen et al., 2021a, 2021b, 2022, 2024; Nguyen and Park, 2023), the time step was set to 0.1% of the undulatory frequency of the body. To model the flow characteristics over the body, while accounting for surface roughness, the k–ε two-equation turbulent model was employed, as it effectively captures the convection and diffusion of turbulent energy. This model is particularly suitable for the flow regimes with the high Re values of 4.8×10^5 for 26-V and 5.3×10^5 for 30-V swimming, which are calculated as follows:

$$Re = \frac{\overline{U}_2 \times L}{\nu}, \tag{F1}$$

where ν is the water kinematic viscosity at 20 °C (1.004×10^{-6} m^2/s), $\overline{U}_2$ represents the averaged swimming speeds of 1.67 m/s at 30 V and 1.56 m/s at 26 V.

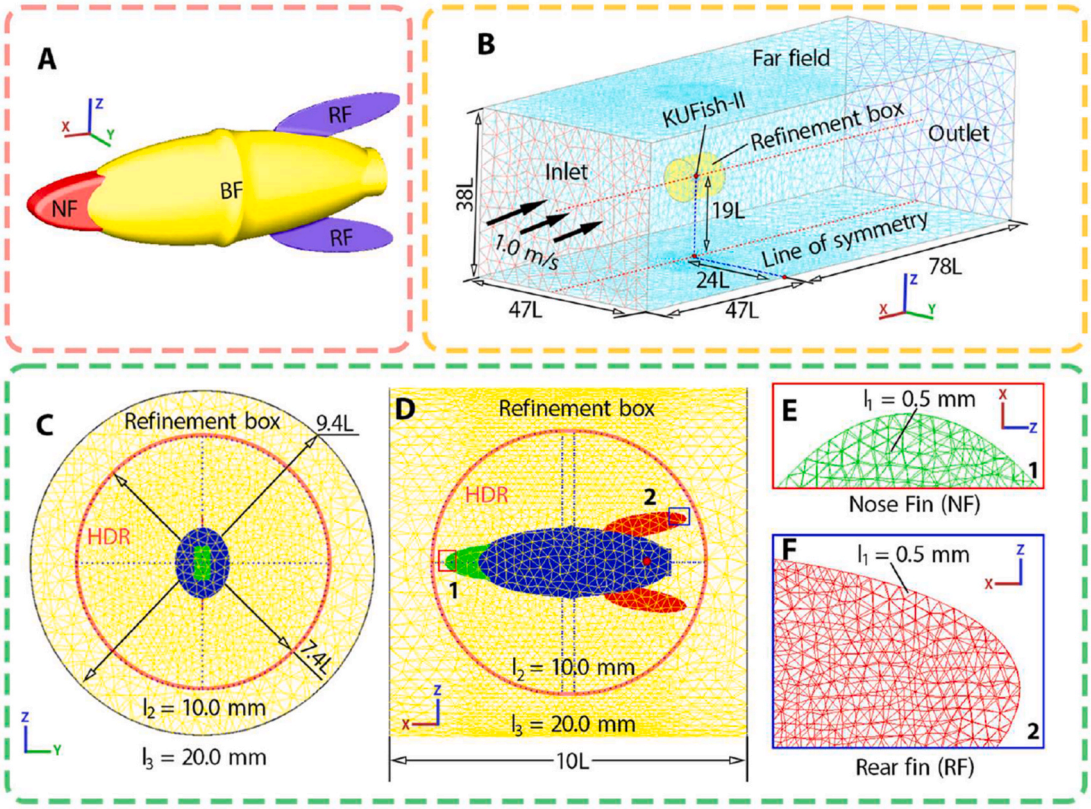


Fig. F1. CFD modeling. (**A**) Outer shape of the robot. (**B-F**) Mesh configurations, including (B) full computational domain; (C) front and (D) side views of the refinement boxes around the body with high-density regions (HDRs); (E) close-up view of the surface mesh at the nose fin and (F) at the rear fin.

F.2. Convergence criteria

The CFD convergence was achieved with residual errors of the order of 10^{-6} for the continuity equation and the k–ε turbulence model, and 10^{-9} for the velocity components. This ensured that the solutions converged with insignificant changes between successive iterations during steady swimming. Similar convergence criteria were obtained for the undulatory swimming simulations. At each timestep, the results reached residuals of 10^{-6} and 10^{-9}, corresponding to the continuity equation and k–ε turbulence model, and the velocity components, respectively.

F.3. Definition of the longitudinal force coefficient

The coefficient of longitudinal force along the x-axis, C_{Fx} is computed as follows:

$$C_{Fx} = \frac{2 \times F_x}{\rho \times S \times \overline{U}_{2'}^2}, \tag{F2}$$

where F_x is the cycle-average force on the x-axis computed using CFD, and $\overline{U}_{2'}$ is the reference speed of 1.0 m/s.

F.4. Mesh convergence study for the simulation of static swimming

A mesh convergence study was conducted on a zero-roughness model over three-body undulatory cycles. The results from the third cycle showed nearly identical values of the cycle-averaged C_{Fx} across the three mesh types: $C_{Fx} = -0.029$ for both coarse and medium meshes and $C_{Fx} = -0.030$ for the fine mesh. Thus, the mesh was considered to converge for the zero-roughness model and was assumed to be sufficiently accurate for simulations with a roughness height of 0.5 mm as well.

Appendix G. Comparison between the actual swimming speed and three predicted values

Table B1 compares the average predicted swimming speed with the measured value of 1.79 m/s. Due to a 6% higher tailbeat frequency (12.2 Hz versus 11.5 Hz), the measured swimming speed was approximately 2.1% higher than the predicted value of 1.75 m/s. In addition, three actual physical parameters of *Fr*, *St*, and σ were compared with their predicted counterparts (see Table B1). The results showed that the actual Fr (1.01) was nearly identical to its prediction (1.00), while the actual σ (13.70) slightly differed by 3% from the predicted value (13.30).

However, a 35% decrease in the actual A_m $(0.13L)$ compared to its predicted value $(0.2L)$ was the primary result of the 33% discrepancy between the actual *St* (0.28) and its predicted value (0.42) (Table B1). The actual A_m was measured using 2D-pixel-based coordinates extracted from the video. For validation, the peak-to-peak distance were also computed using the measured amplitudes of the undulatory angles of the body, peduncle, and CF (as shown in Fig. 5C). The computed A_m was also approximately $0.13L - 0.14L$, closely matching the pixel-based measurement (Fig. G1). Thus, properly estimating A_m is important for the initial prediction of the target swimming speed.

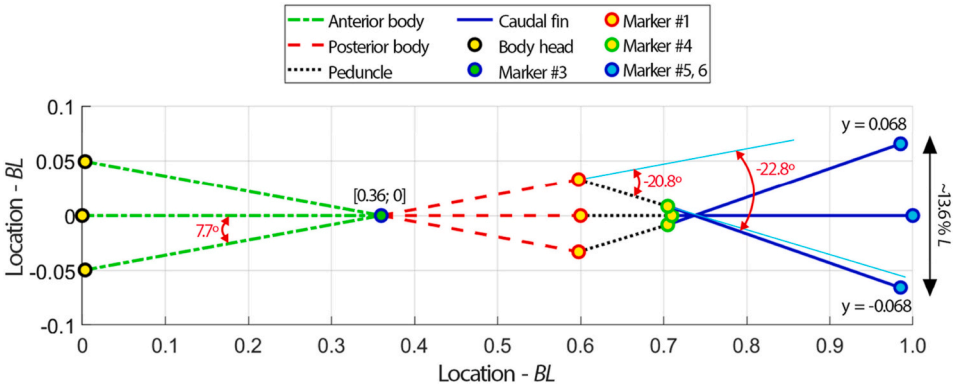


Fig. G1. Validation of the measured pixel-based peak-to-peak amplitude. Since the phase of body undulation was opposite to those of the peduncle and CF, the calculation used the maximum undulatory angles for body (± 7.7°), peduncle (∓ 20.8°), and CF (∓ 22.8°), as shown in Fig. 5C. Notably, the undulatory frequencies of the body, peduncle, and CF were assumed to be identical.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.oceaneng.2026.128377.

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